

KCCITM NOIDA • AKTU

WEEK 2 MODULE

PROGRAMMING PRACTICE



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SUBMISSION DATE

19 July 2026

Module 2

1) Fraud Transaction detection [linear search]

code:

```
Public Class Fraud Transaction Detection {  
    Public static void main (String [] args) {  
        Scanner sc = new Scanner (System.in);  
        int n = sc.nextInt();  
        int [] transaction = new int [n];  
        for (int i = 0; i < n; i++) {  
            transaction [i] = sc.nextInt();  
        }  
        int suspicious ID = sc.nextInt();  
        boolean flag = false;  
        for (int i : transaction) {  
            if (i == suspicious ID) {  
                flag = true;  
                break;  
            }  
        }  
        if (flag) {  
            System.out.println ("Fraud Transaction found");  
        } else {  
            System.out.println ("Fraud Transaction Not found");  
        }  
        sc.close();  
    }  
}
```



```
collegeTraining Version control FraudTransactionDetection
Week1Module.java FraudTransactionDetection.java
5 public class FraudTransactionDetection {
6     public static void main(String[] args) {
7         Scanner sc = new Scanner(System.in);
8         int n = sc.nextInt();
9         int[] transaction = new int[n];
10        for (int i = 0; i < n; i++) {
11            transaction[i] = sc.nextInt();
12        }
13        int suspiciousID = sc.nextInt();
14        boolean flag = false;
15        for (int i: transaction){
16            if (i == suspiciousID){
17                flag = true;
18                break;
19            }
20        }
21        if(flag){
22            System.out.println("Fraud Transaction Found");
23        }else{
24            System.out.println("Fraud Transaction Not Found");
25        }
26        sc.close();
27    }
28 }
```

Run FraudTransactionDetection

```
5
1201 1305 1450 1408 1502
1450
Fraud Transaction Found
```

2. Server log search [binary search]

```
public class server log search {  
    public static void main (String [] args) {  
        Scanner sc = new Scanner (System.in);  
        int n = sc.nextInt();  
        int [] logs = new int [n];  
        for (int i = 0; i < n; i++) {  
            logs[i] = sc.nextInt();  
        }  
        int target = sc.nextInt();  
        boolean found = false;  
        int start = 0;  
        int end = n - 1;  
        while (start <= end) {  
            int mid = start + (end - start) / 2;  
            if (logs[mid] == target) {  
                found = true;  
                break;  
            } else if (logs[mid] < target) {  
                start = mid + 1;  
            } else {  
                end = mid - 1;  
            }  
        }  
        if (found) {  
            System.out.println ("log found");  
        } else {  
            System.out.println ("log not found");  
        }  
    }  
}
```

collegeTrainingVersion controlServerLogSearch

Week1Module.javaFraudTransactionDetection.javaServerLogSearch.java

```
5 public class ServerLogSearch {
6     public static void main(String[] args) {
7         Scanner sc = new Scanner(System.in);
8         int n = sc.nextInt();
9         int[] logs = new int[n];
10        for (int i = 0; i < n; i++) {
11            logs[i] = sc.nextInt();
12        }
13        int target = sc.nextInt();
14        boolean found = false;
15        int start = 0;
16        int end = n - 1;
17        while (start <= end) {
18            int mid = start + (end - start) / 2;
19            if (logs[mid] == target) {
20                found = true;
21                break;
22            } else if (logs[mid] < target) {
23                start = mid + 1;
24            } else {
25                end = mid - 1;
26            }
27        }
28    }
29 }
```

RunServerLogSearch

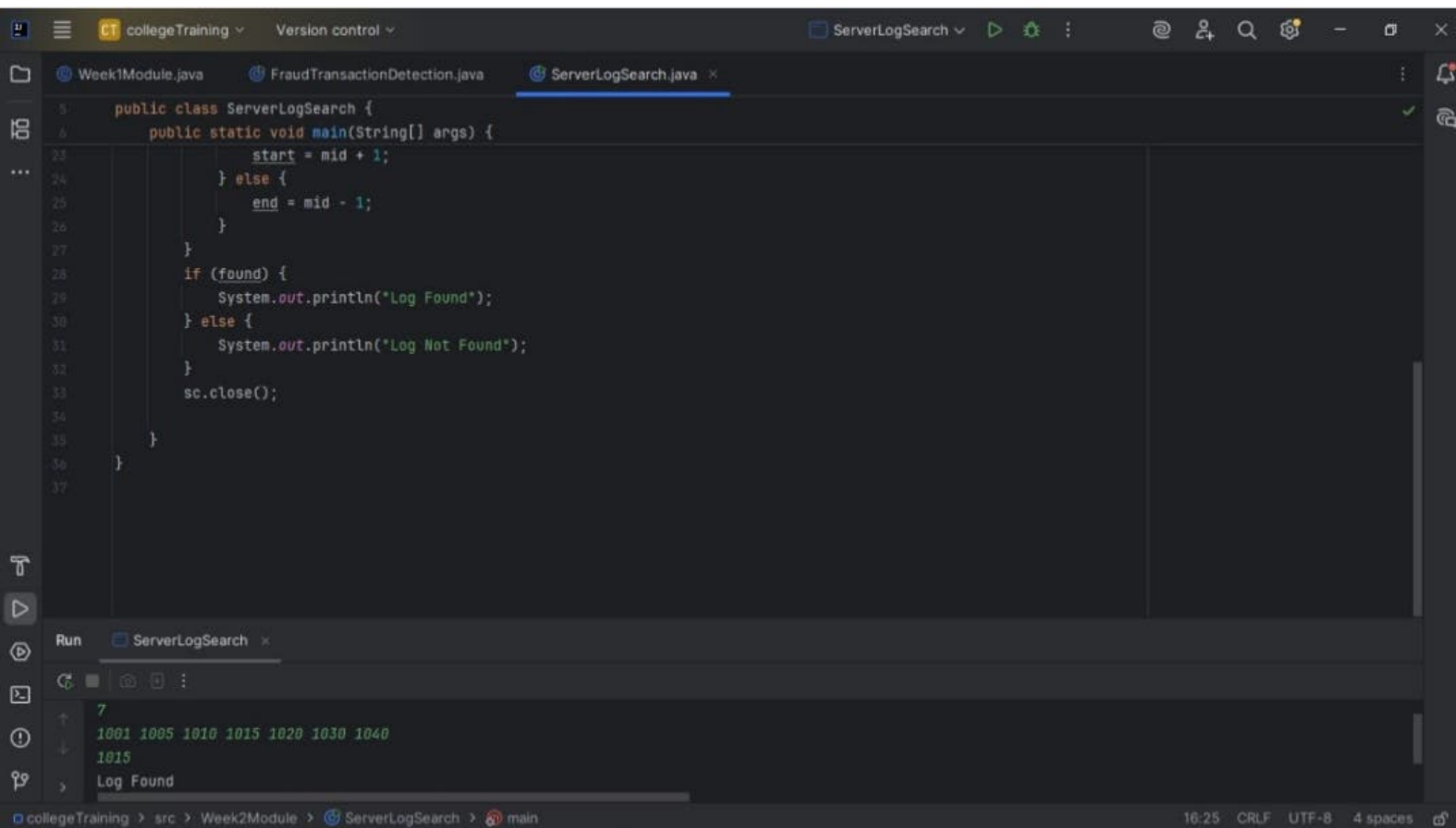
7

1001 1005 1010 1015 1020 1030 1040

1015

Log Found

collegeTraining > src > Week2Module > ServerLogSearch > main16:25 CRLF UTF-8 4 spaces



3. Sort employee salaries [bubble sort]

```
Public class sort salaries {
```

```
    public static void main (String[] args) {
```

```
        Scanner sc = new Scanner (System.in);
```

```
        int n = sc.nextInt();
```

```
        int[] salaries = new int[n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            salaries[i] = sc.nextInt();
```

```
        }
```

```
        for (int i = 0; i < n - 1; i++) {
```

```
            for (int j = 0; j < n - i - 1; j++) {
```

```
                if (salaries[j] > salaries[j+1]) {
```

```
                    int temp = salaries[j];
```

```
                    salaries[j] = salaries[j+1];
```

```
                    salaries[j+1] = temp;
```

```
                }
```

```
            }
```

```
        }
```

```
        for (int i : salaries) {
```

```
            System.out.print(i + " ");
```

```
        }
```

```
        sc.close();
```

```
    }
```

```
}
```

```
public class SortSalaries {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] salaries = new int[n];
        for (int i = 0; i < n; i++) {
            salaries[i] = sc.nextInt();
        }
        for (int i = 0; i < n-1; i++) {
            for (int j = 0; j < n-i-1; j++) {
                if(salaries[j] > salaries[j+1]){
                    int temp = salaries[j];
                    salaries[j] = salaries[j+1];
                    salaries[j+1] =temp;
                }
            }
        }
        for(int i : salaries){
            System.out.print(i + " ");
        }sc.close();
    }
}
```

Run SortSalaries

```
5
45000 32000 55000 28000 60000
28000 32000 45000 55000 60000
Process finished with exit code 0
```


4. Sort website Response Time (Insertion sort)

```
public class sort Response Time {
```

```
    public static void main (String [] args) {
```

```
        Scanner sc = new Scanner (System.in);
```

```
        int n = sc.nextInt();
```

```
        int [] responseTimes = new int [n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            responseTimes [i] = sc.nextInt();
```

```
        }
```

```
        for (int i = 1; i < n; i++) {
```

```
            int key = responseTimes [i];
```

```
            int j = i - 1;
```

```
            while (j >= 0 && responseTimes [j] > key) {
```

```
                responseTimes [j+1] = responseTimes [j];
```

```
                j--;
```

```
            }
```

```
            responseTimes [j+1] = key;
```

```
        }
```

```
        for (int i : responseTimes) {
```

```
            System.out.print (i + " ");
```

```
        }
```

```
        sc.close();
```

```
    }
```

```
}
```

```
collegeTraining Version control SortResponseTime
Week1Module.java FraudTransactionDetection.java ServerLogSearch.java SortSalaries.java SortResponseTime.java
5 public class SortResponseTime {
6     public static void main(String[] args) {
7         Scanner sc = new Scanner(System.in);
8         int n = sc.nextInt();
9         int[] responseTimes = new int[n];
10        for (int i = 0; i < n; i++) {
11            responseTimes[i] = sc.nextInt();
12        }
13        for (int i = 1; i < n; i++) {
14            int key = responseTimes[i];
15            int j = i-1;
16            while(j >= 0 && responseTimes[j] > key){
17                responseTimes[j+1] = responseTimes[j];
18                j--;
19            }
20            responseTimes[j+1] = key;
21        }
22        for(int i: responseTimes){
23            System.out.print(i + " ");
24        }
25        sc.close();
26    }
27 }
Run SortResponseTime
5
250 120 320 180 100
100 120 180 250 320
collegeTraining > src > Week2Module > SortResponseTime > main 25:20 CRLF UTF-8 4 spaces
```

5. Prioritize Bug Reports (selection sort)

```
Public class BugReports {
```

```
    public static void main (String [] args) {
```

```
        Scanner sc = new Scanner (System.in);
```

```
        int [] priorityscore = new int [n];
```

```
        for (int i = 0; i < n; i++) {
```

```
            priorityscore [i] = sc.nextInt();
```

```
        }
```

```
        for (int i = 0; i < n-1; i++) {
```

```
            int min = i;
```

```
            for (int j = i+1; j < n; j++) {
```

```
                if (priorityscore [j] < priorityscore [min]) {
```

```
                    min = j;
```

```
                }
```

```
            }
```

```
            int temp = priorityscore [i];
```

```
            priorityscore [i] = priorityscore [min];
```

```
            priorityscore [min] = temp;
```

```
        }
```

```
        for (int i : priorityscore) {
```

```
            System.out.print (i + " ");
```

```
        } sc.close();
```

```
    }
```

```
}
```


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Version control

BugReports

Run

Search

Run and Debug

Run and Test

Run and Profile

Run and Benchmark

Run and Measure

Run and Monitor

Run and Log

Run and Trace

Run and Profile

Run and Benchmark

Run and Measure

Run and Monitor

Run and Log

Run and Trace

Week1Module.java

FraudTransactionDetection.java

ServerLogSearch.java

SortSalaries.java

SortResponseTime.java

BugReports.java

```
5 public class BugReports {
6     public static void main(String[] args) {
7         Scanner sc = new Scanner(System.in);
8         int n = sc.nextInt();
9         int[] priorityScore = new int[n];
10        for (int i = 0; i < n; i++) {
11            priorityScore[i] = sc.nextInt();
12        }
13        for (int i = 0; i < n-1; i++) {
14            int min = i;
15            for (int j = i+1; j < n; j++) {
16                if(priorityScore[j] < priorityScore[min])
17                    min = j;
18            }
19            int temp = priorityScore[i];
20            priorityScore[i] = priorityScore[min];
21            priorityScore[min] = temp;
22        }
23        for(int i : priorityScore){
24            System.out.print(i + " ");
25        }
26        sc.close();
27    }
28 }
```

Run

BugReports

5

9 2 8 1 5

1 2 5 8 9

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src

Week2Module

BugReports

main

16:59

CRLF

UTF-8

4 spaces